



3ª Série

Matemática

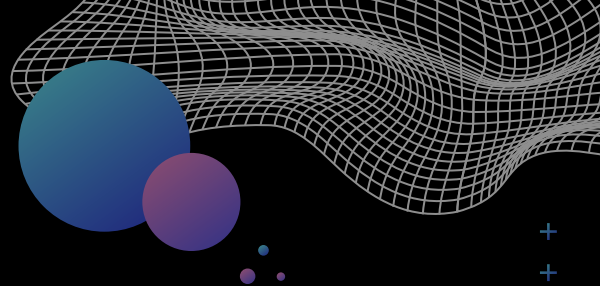
Aula 05 - 2º Bim

Prof. Diogo Pelaes

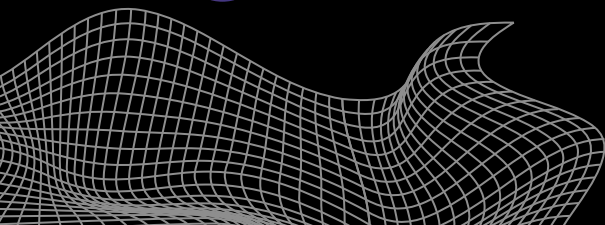
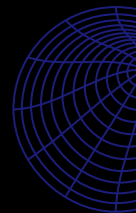


Distância entre ponto e reta

- Fazer exercícios do 1 ao 8 (só nesse arquivo).



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Distância entre ponto e reta

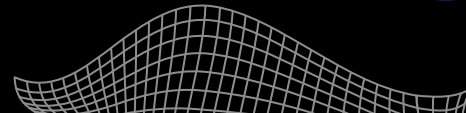
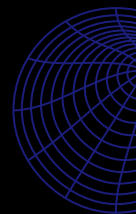
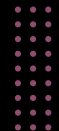
A **distância** d entre um ponto $P(x_0, y_0)$ e uma reta r de equação geral $ax + by + c = 0$ é dada por:

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$



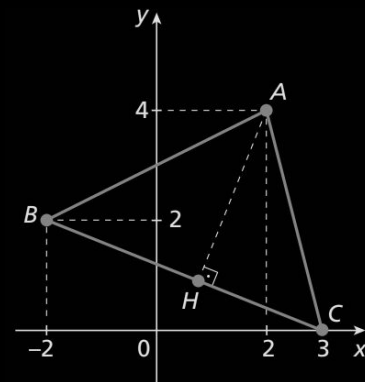
Exemplo

Vamos aplicar esse teorema no cálculo da distância d entre o ponto $P(1, 5)$ e a reta r de equação $y = x - 2$.



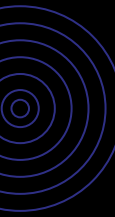
Exercício Resolvido

Dado o triângulo de vértices $A(2, 4)$, $B(-2, 2)$ e $C(3, 0)$, calcular a medida de sua altura relativa ao lado $\overline{BC}$.

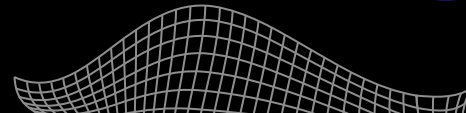
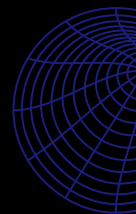
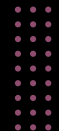




Exercício Resolvido



Calcular a medida da distância entre as retas paralelas r e s , de equações $2x - y + 4 = 0$ e $2x - y - 7 = 0$, respectivamente.





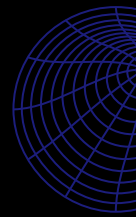
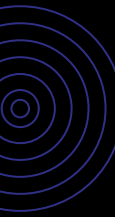
1. Calcule a distância do ponto P à reta r em cada um dos casos a seguir.

a. $P(1, 3)$ e $(r) 5x + 12y - 2 = 0$

b. $P(-2, -4)$ e $(r) y = x - 8$

c. $P(9, 6)$ e $(r) x = 4$

d. $P(5, -2)$ e $(r) y = 3$

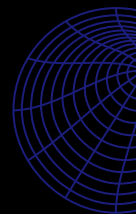
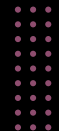
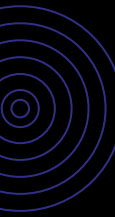


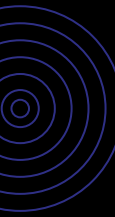
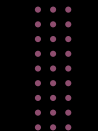
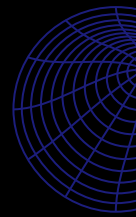


2. Os vértices de um triângulo são dados pelos pontos $A(1, 3)$, $B(5, 0)$ e $C(0, 5)$.

a. Obtenha uma equação geral da reta $\overleftrightarrow{AB}$.

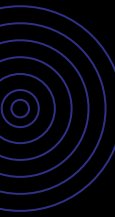
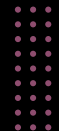
b. Calcule a medida da altura relativa ao vértice C .

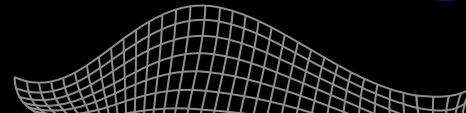
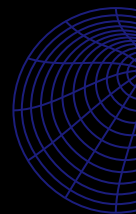


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3. Um triângulo tem vértices $A(2, 0)$, $B(3, 1)$ e $C(0, 2)$. Calcule a medida da altura do triângulo relativa ao lado $\overline{BC}$
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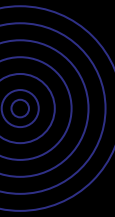
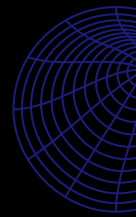
4. Dadas as retas $r: 2x + 5y - 4 = 0$ e $s: 5x - 2y + 8 = 0$, determine as equações das retas cujos pontos são equidistantes de r e s . Siga estes passos:

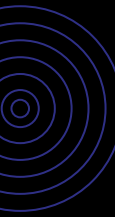
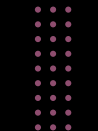
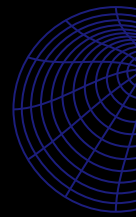
- considere um ponto genérico $P(x, y)$;
 - use a definição $d_{P, r} = d_{P, s}$;
 - obtenha as equações das retas eliminando os módulos.
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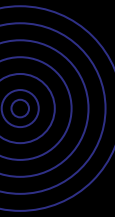
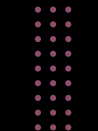
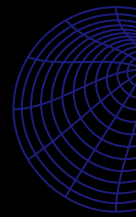


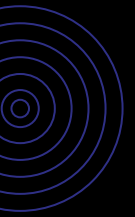


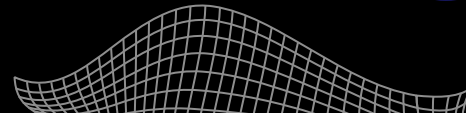
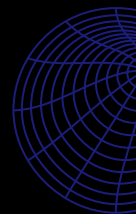
5. Considerando a reta r de equação $4x - 3y - 12 = 0$, determine:

- a. os pontos do eixo das abscissas que distam 4 unidades de r ;
 - b. os pontos da bissetriz dos quadrantes ímpares que distam 2 unidades de r ;
 - c. os pontos que pertencem à reta s de equação $y = x + 2$ e distam 3 unidades de r .
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6. No plano cartesiano xOy , a reta t , representada pela equação $y = -\frac{3x}{4}$, tangencia um círculo de centro $C(3, 4)$. Obtenha a área A desse círculo.
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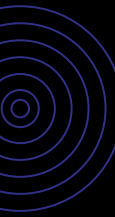
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7. Obtenha a medida da distância, da origem do plano cartesiano à reta de equação $3x + 4y - 4 = 0$.
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8. Obtenha a medida da distância entre as retas paralelas
 $2x - 3y + 5 = 0$ e $4x - 6y - 1 = 0$.





Gabrito



1. a. 3 b. $3\sqrt{2}$ c. 5 d. 5

2. a. $3x + 4y - 15 = 0$ b. 1 c. $\frac{5}{2}$

3. $\frac{2\sqrt{10}}{5}$

4. $3x - 7y + 12 = 0$ e $7x + 3y + 4 = 0$.



5. a. $A(8, 0)$ e $A'(-2, 0)$
b. $B(22, 22)$ e $B'(2, 2)$
c. $C(33, 35)$ e $C'(3, 5)$

6. 25π

7. $\frac{4}{5}$

8. $\frac{11\sqrt{13}}{26}$

